Advanced Data Management

Course Introduction

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Introduction

Course Objectives

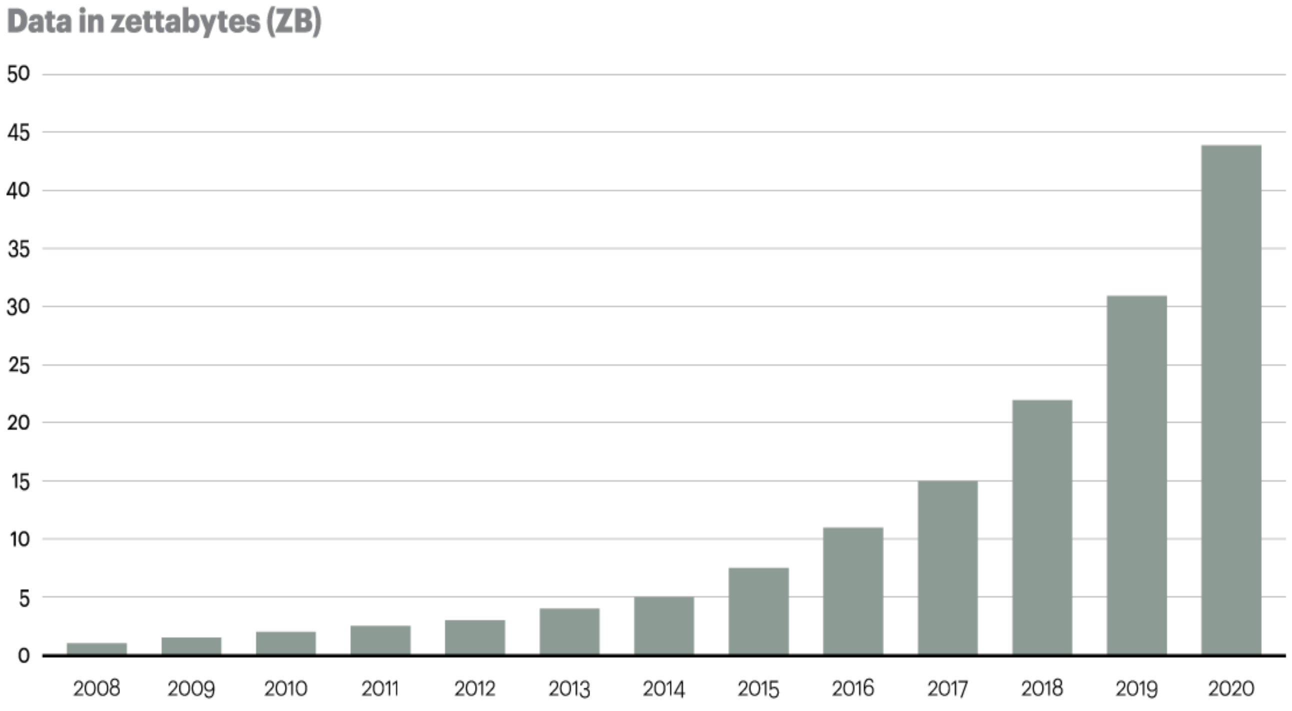
1. Provide the basics of information integration

2. Develop a rigorous theory of data integration through logics

Why?

Data is Ever-Growing!

Data is growing at a 40 percent compound annual rate, reaching nearly 45 ZB by 2020



Source: Oracle, 2012

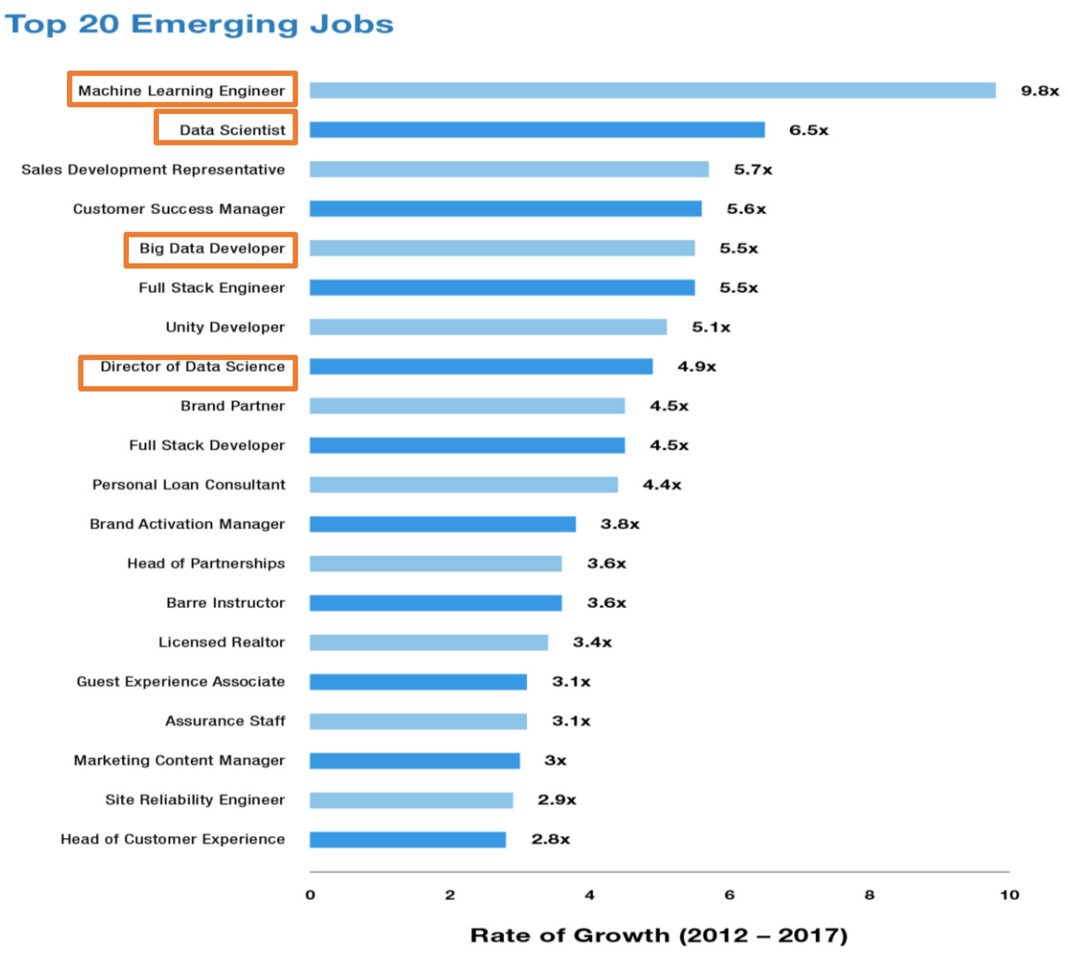
Source: http://www.atkearney.com/

WHAT'S A ZETTABYTE?

1 kilobyte	1,000
1 megabyte	1,000,000
1 gigabyte	1,000,000,000
1 terabyte	1,000,000,000,000
1 petabyte	1,000,000,000,000,000
1 exabyte	1,000,000,000,000,000,000
1 zettabyte	1,000,000,000,000,000,000,000

SOURCES: CISCO

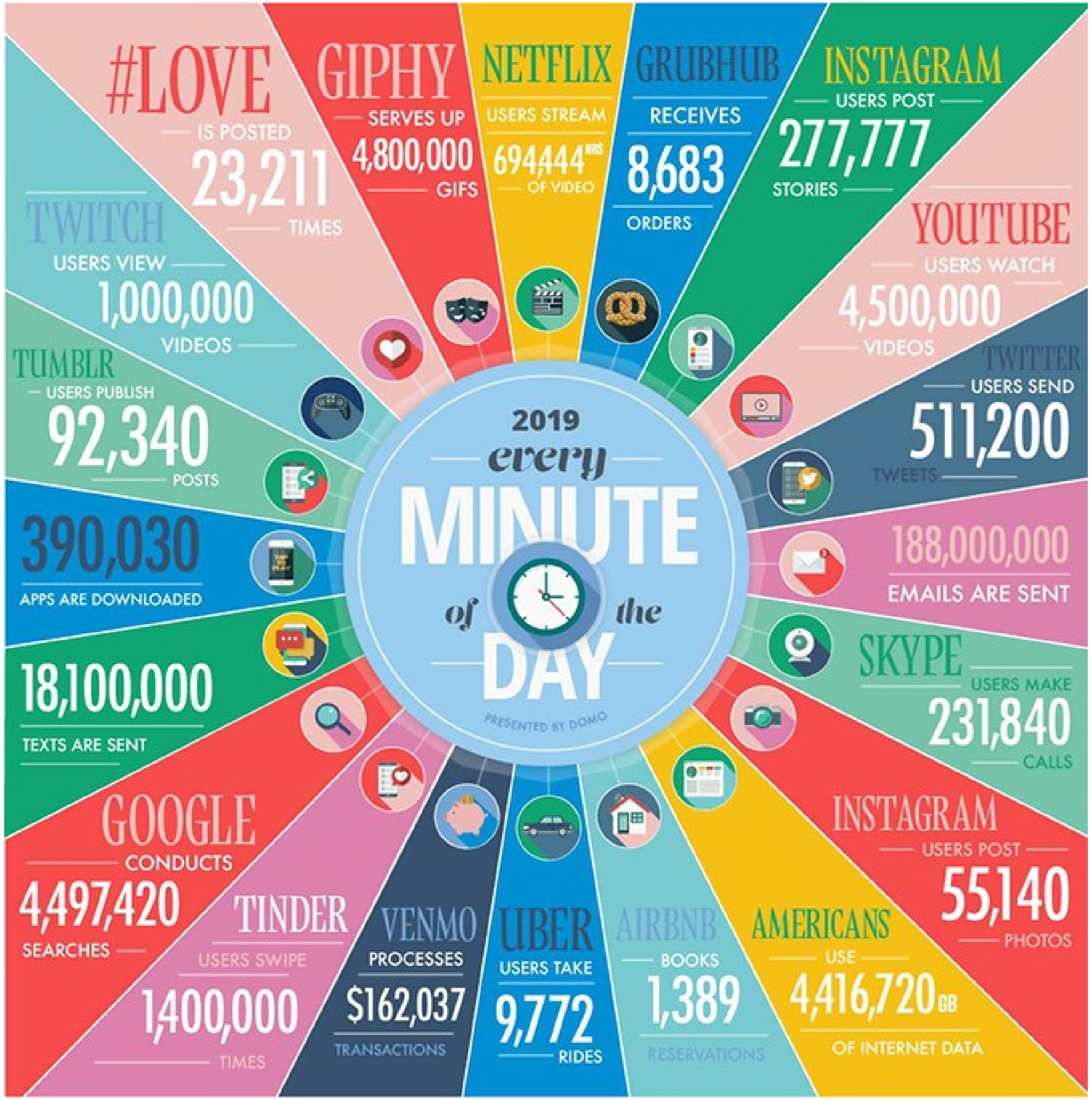
The Data Market is Ever-Growing



Who is producing data

<u>Who is producing all this data?</u>

1. Business Information Systems
2. Social media
3. Sensors and IoT
4. **US!**


2019 every MINUTE of the DAY
PRESENTED BY DOMO
#LOVE IS POSTED 23,211 TIMES
GIPHY SERVES UP 4,800,000 GIFS
NETFLIX USERS STREAM 694,444 HRS OF VIDEO
GRUBHUB RECEIVES 8,683 ORDERS
INSTAGRAM USERS POST 277,777 STORIES
YOUTUBE USERS WATCH 4,500,000 VIDEOS
TWITTER USERS SEND 511,200 TWEETS
188,000,000 EMAILS ARE SENT
SKYPE USERS MAKE 231,840 CALLS
INSTAGRAM USERS POST 55,140 PHOTOS
AMERICANS USE 4,416,720 GB OF INTERNET DATA
AIRBNB BOOKS 1,389 RESERVATIONS
UBER USERS TAKE 9,772 RIDES
VENMO PROCESSES $162,037 TRANSACTIONS
TINDER USERS SWIPE 1,400,000 TIMES
GOOGLE CONDUCTS 4,497,420 SEARCHES
18,100,000 TEXTS ARE SENT
390,030 APPS ARE DOWNLOADED
TUMBLR USERS PUBLISH 92,340 POSTS
TWITCH USERS VIEW 1,000,000 VIDEOS

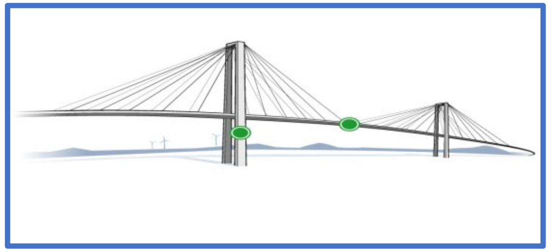
Pervasive Computing

Lots of data is produced by machines

DOMOTICS

SMART LIGHTNING SYSTEM

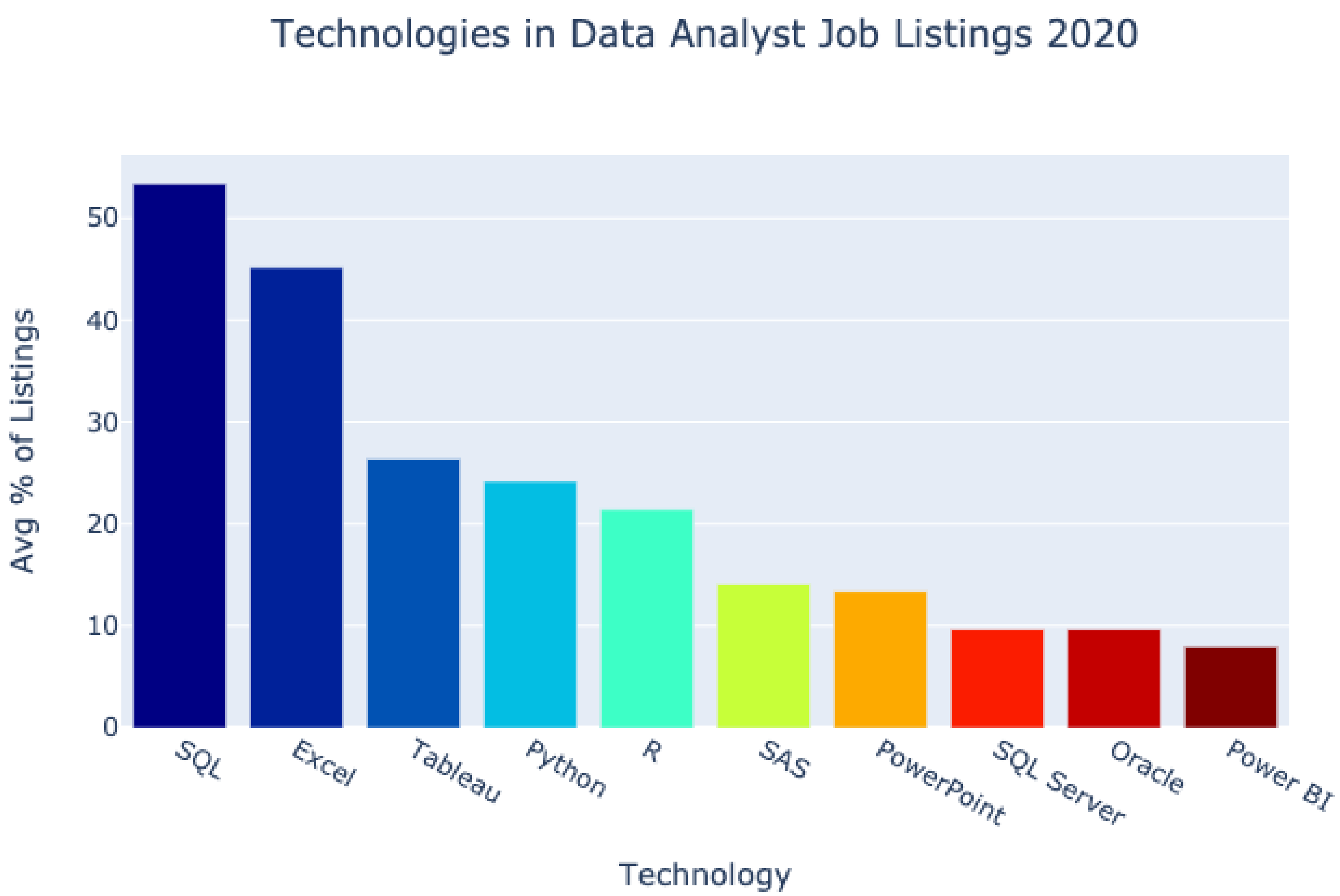
STRUCTURAL MONITORING



SMARTWATCHES

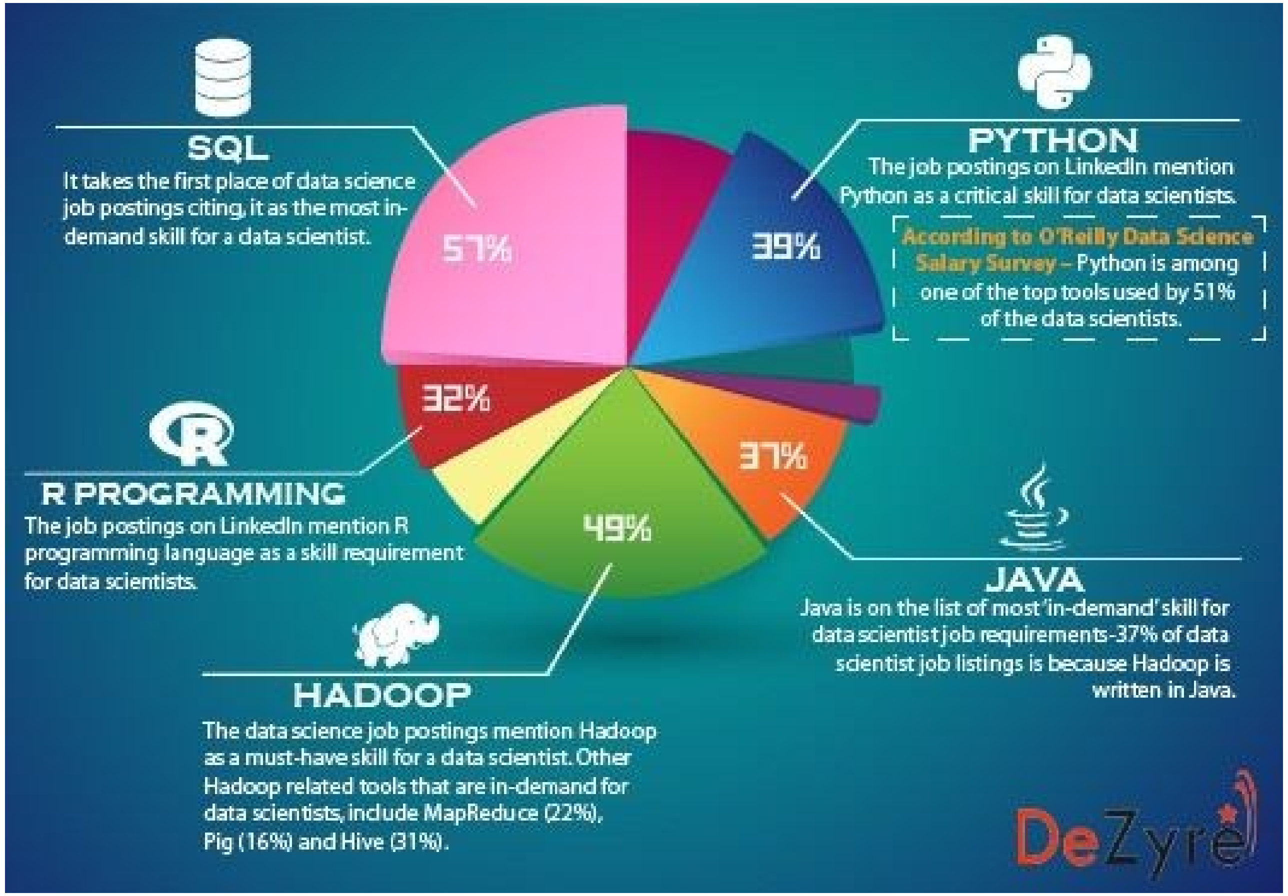
CARS

Structured Data Manipulation!



Structured Data Manipulation!

Data **Scientist** Skills in Job Ads



Structured Data Manipulation!

Why structured data manipulation is so important?

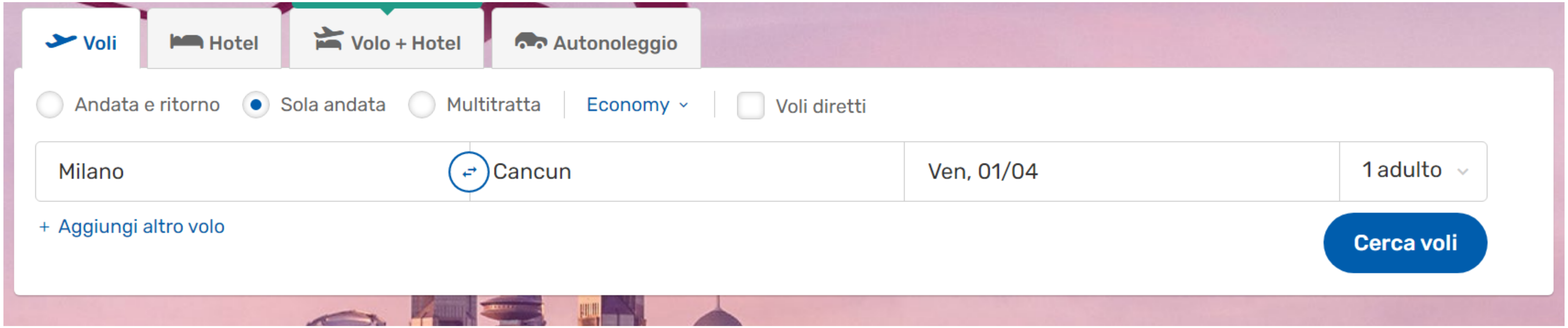
Structured Data Manipulation!

It allows companies and organizations to **effectively access their data** and commutate it into the **information** that is **critical to their business and aims**.

Structured data manipulation process is also a valuable tool in **identifying data inconsistencies and redundancies**.

Enable **valuable functionalities and services** for companies and their customers\users.

Example: Escape the Advanced Data Management Course!



1. Open the list of destinations (Destinations available from different websites)
2. Open the list of available dates (Dates available from different websites)
3. Search…

…Where does the data come from?

Example: Escape the Advanced Data Management Course!

- Where do flight options come from?
  - Different sources: companies, brokers…
  - Different entities may represent data (flights) in different ways

- Are the flight options still available?
  - Your colleagues may have booked all the flights already!
  - Data may have been pre-loaded and not up to date.

- Confirm booking.
  - The escape is set!
  - You produce more data (your seat is no longer available).

Example: Escape the Advanced Data Management Course!

- Where do flight options come from?
  - Different sources: companies, brokers…
  - Different entities may represent data (flights) in different ways
- Are the
  - Your
  - Data
- Confirm
  - The escape is set!
  - You produce more data (your seat is no longer available).

We need integrating information from multiple sources

Integrating Information from Multiple Sources

- Integrating information from multiple sources is **challenging**

- Data may reside in **different physical** places
  - Load a snapshot? (**materialization**)
  - Keep hitting the sources with queries? (**virtualization**)

- Different sources may represent data in different ways (**heterogeneity**)
  - Physically, data may be stored in different formats
  - Conceptually, data may represent different aspects of the same reality.

The goal of information integration is to provide the user with a reconciled view of data stored in different sources

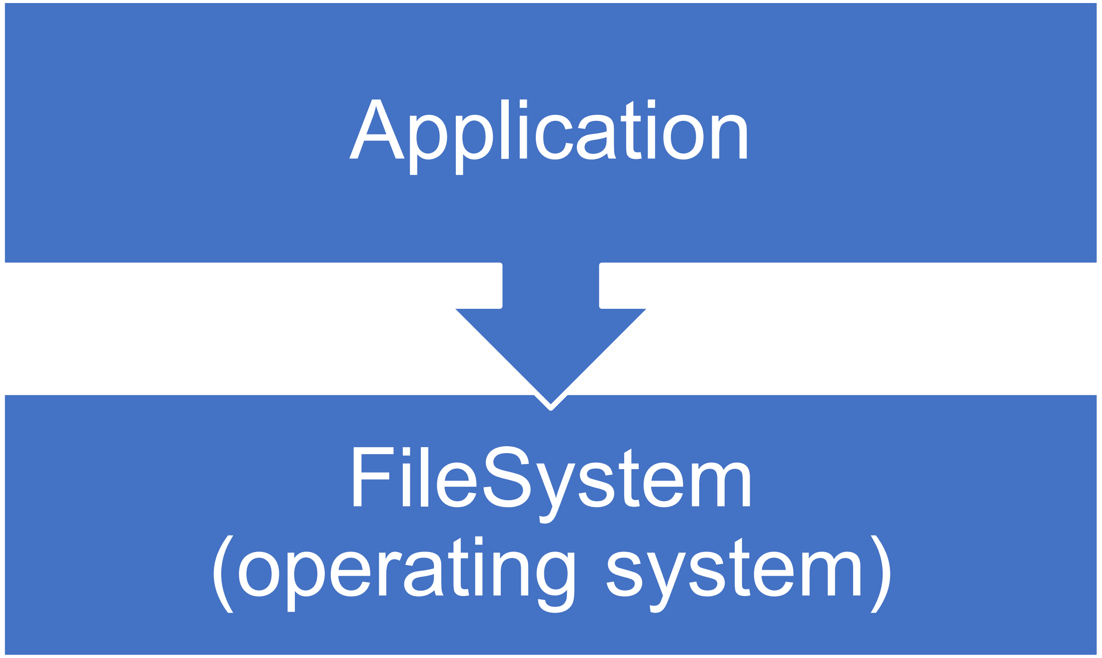
Data Heterogeneity

Physical Heterogeneity

- Information may be stored in different formats
  - Depending on the technology used

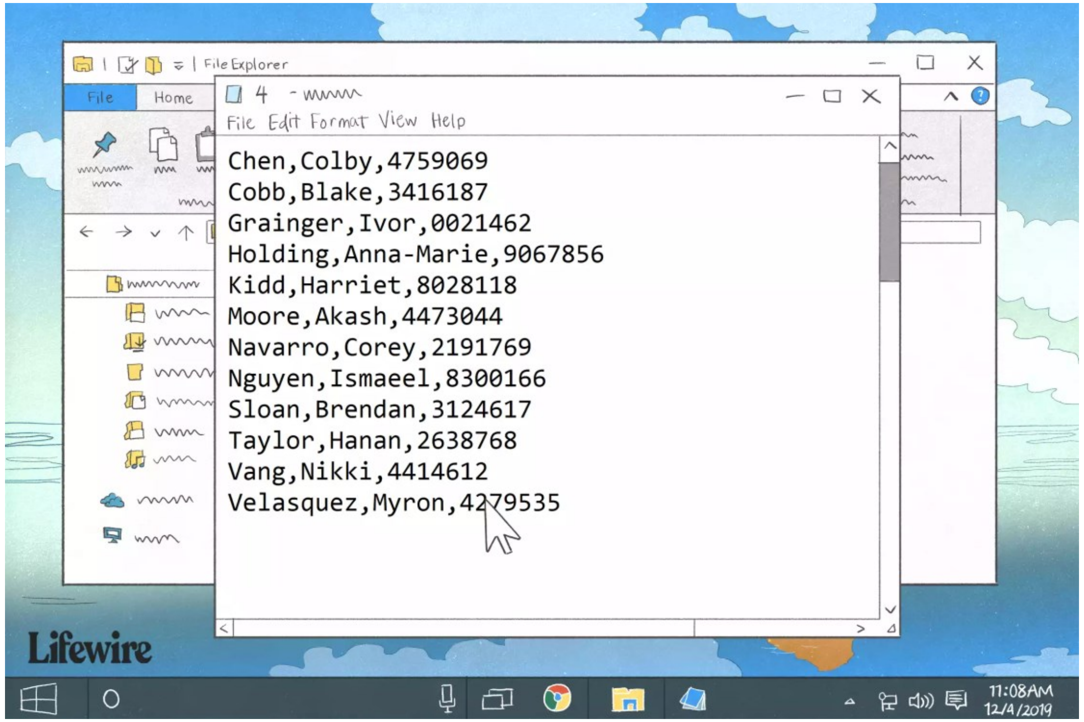
File-Based Approach

- Applications manage information using the OS file system via APIs



File-Based Approach

- The information in the files may take different shapes
- **Structured**.
  - Information takes a tabular shape.
  - **Example.** Comma Separated Values (CSV) files



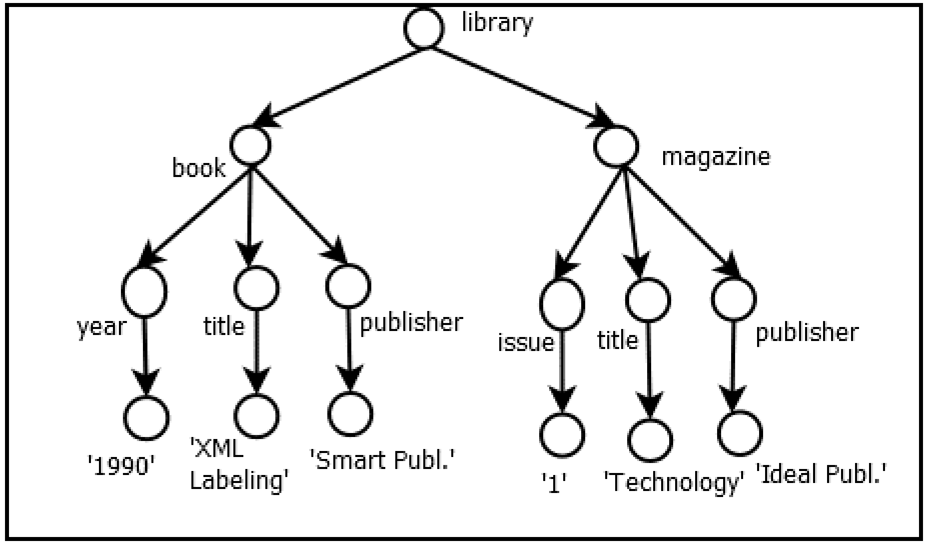
File-Based Approach

- The information in the files may take different shapes
- **Semi-Structured**.
  - Information has a shape, but there is more flexibility.
  - **Example.** eXtensible Markup Language (XML) files

a) XML document

```
<library>
   <book>
        <year>1990</year>
        <title>XML Labeling</title>
        <publisher>Smart Publ.</publisher>
   </book>
   <magazine>
          <issue>1</issue>
          <title>Technology</title>
          <publisher>Ideal Publ.</publisher>
   <magazine>
</library>
```

b) XML Tree representation



File-Based Approach

- The information in the files may take different shapes
- **Unstructured**.
  - Information has no predefined structure.
  - **Example.** Text (TXT) files
  - Hard to interpret…

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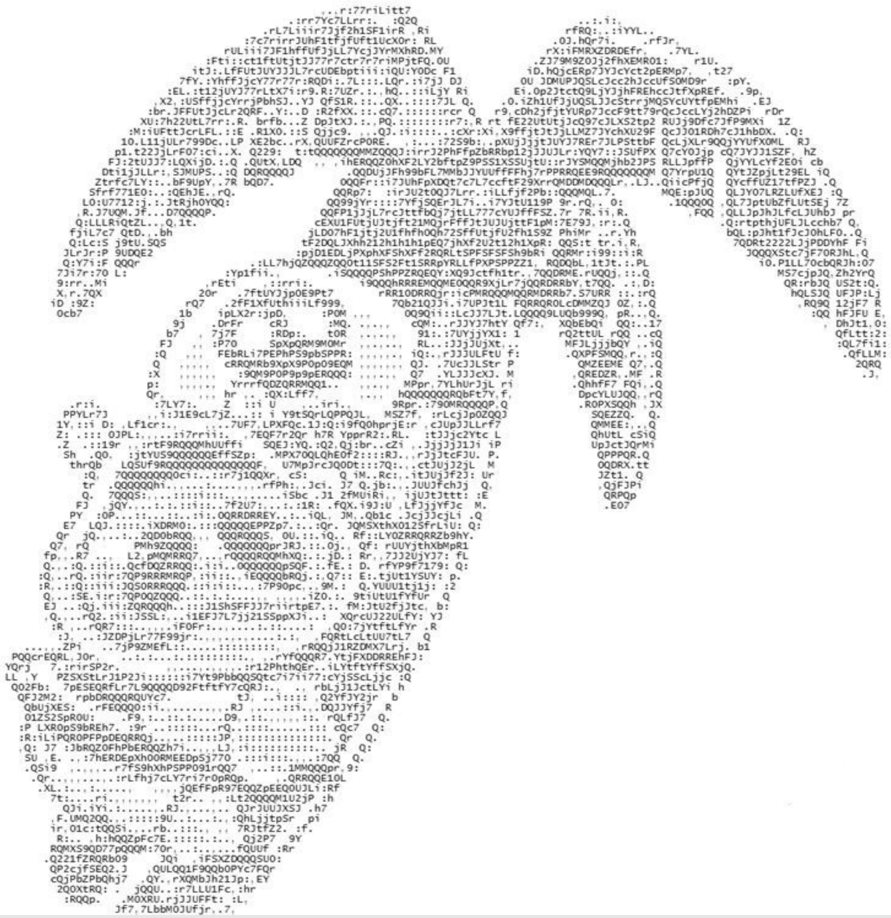


File-Based Storage: Examples

1. XML Files
2. Excel sheets
3. CSV files.
4. Text

A More Structured Approach.

- In many cases, we cannot rely only on files.
- If the size of the data is big (GBs,TBs…)
  - We need efficient mechanisms to store and access
  - Implementing the right algorithm to navigate the data in our application may not be easy
- If multiple concurrent accesses are possible
  - We need to manage concurrent updates
  - Algorithms for that may be tricky to implement.
- In all these cases, the management of data should be left to a specialized piece of software

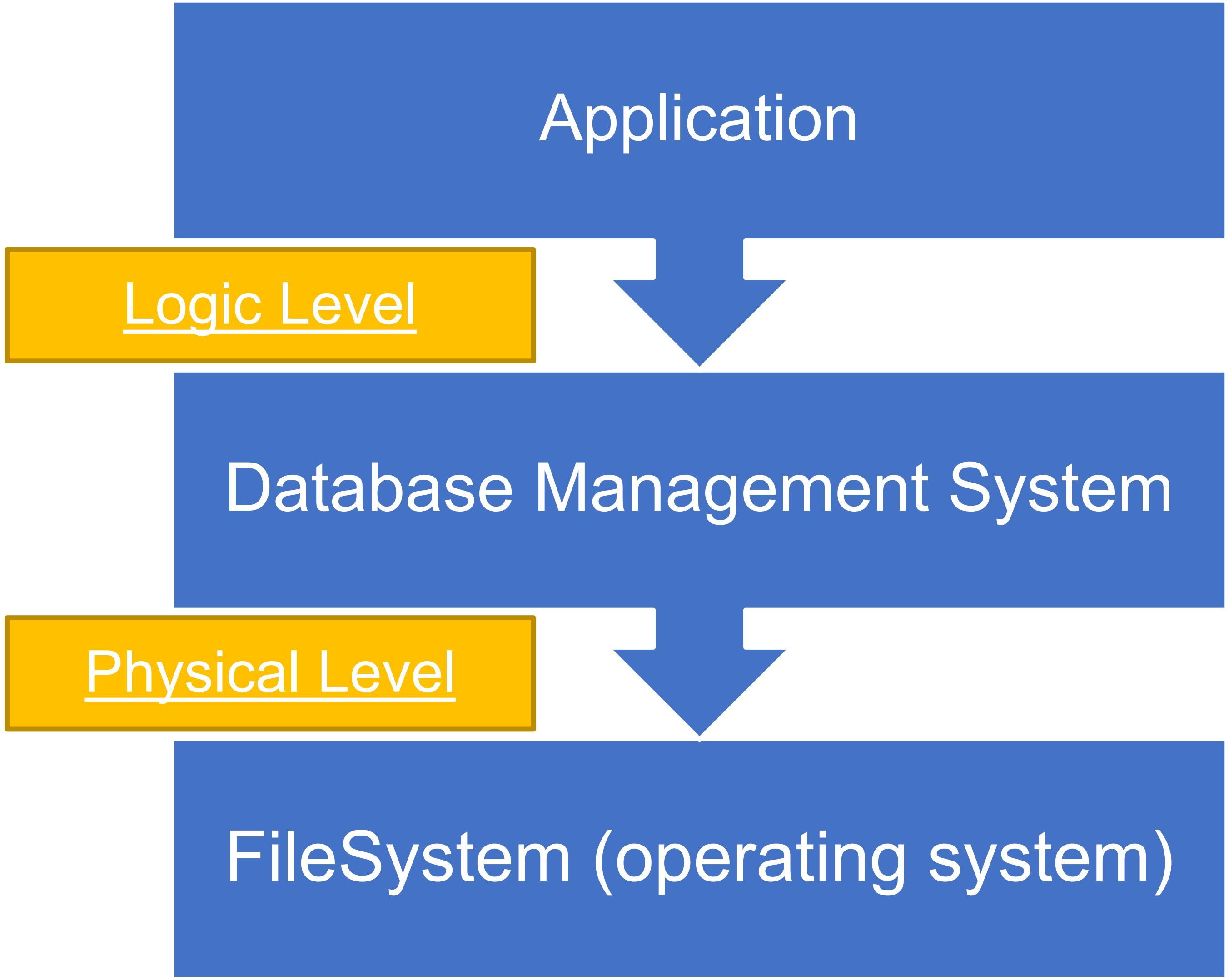
Database Management Systems

- In these cases we use **D**ata**B**ase **M**anagement **S**ystem
  - Specialized piece of software that handles data

- Using DBMSs, the application and data layer are separate
  - Applications interact with the data at the logical level while the physical level is handled by the DBMS

- DBMSs do not solve the information integration problem
  - Different DBMSs may use different data model
  - Different DBSMs may represent data in different way

Application logic / data separation



Different Data Models

- **Legacy databases**
  - Hierarchical model
  - Network model
- **SQL database.**
  - We can use SQL or a dialect thereof.
  - Relational Data Model.
- **Post-Relational**
  - Object Databases
  - Multivalue DBs
  - Graphs
- **NoSQL Databases**
  - Key-value stores
  - Big Tables
  - Documents (Json or the like)

Conceptual Heterogeneity

Information may be represented in different ways from the conceptual point of view.

Conceptual Heterogeneity: Example

Name	DOB	Residency
Amber A.	10/20/2030	New Amsterdam
Rose B.	10/20/1930	Constantinople

ID	Customer Name	Product
O001	Amber A.	P001
O002	Dylan B.	P001

Both databases are relational.

- What are the persons?
- What’s their DOB?

Conceptual Heterogeneity: Example

Person	DOB
Amber A	10/20/2030
Rose B.	10/20/1930
Dylan B.	**Unknown**

Conceptual Heterogeneity: Example

Person	DOB
Amber A	**10/20/2030**
Rose B.	10/20/1930
Dylan B.	**10/20/2030**

Conceptual Heterogeneity: Example

Person	DOB
Amber A.	10/20/2030
Amber A.	**Unknown**
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Dylan B.	**Unknown**

Conceptual Heterogeneity: Example

Person	DOB
Amber A.	10/20/2030
Rose B.	10/20/1930
Dylan B.	**Unknown**
Abraham L.	10/20/1930

Information Management

Data Management

We may want to perform different tasks with the reconciled information:

- **Query:** Answer database queries over the different sources
- **Update:** Change information at the sources
- **Delete:** Remove information from the sources
- **Insert:** Add information into the sources

Source

Name	DOB	Residency
Amber A.	10/20/2030	New Amsterdam
Rose B.	10/20/1930	Constantinople

ID	Customer Name	Product
O001	Amber A.	P001
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Query

- How do we extract information from the sources?

- **Ex: Extract all persons in the data with their Date of Birth (if known).**

Materialization

Person	DOB
Amber A.	10/20/2030
Rose B.	10/20/1930
Dylan B.	**Unknown**

Virtualized

Name	DOB	Residency
Amber A.	10/20/2030	New Amsterdam
Rose B.	10/20/1930	Constantinople

ID	CName	Product
O001	Amber A.	P001
O002	Dylan B.	P001

```
SELECT Name, DOB
FROM Persons
```

```
SELECT CName
FROM Customers
```

Virtual vs Materialized

- **Virtual PROs.**
  - Information is always fresh.
  - No need for extra space.
- **Virtual CONs.**
  - Usually, computationally heavy.
  - Multiple accesses to sources
- **Materialized PROs.**
  - Efficiency (after materialization is done).
  - Single access to the data sources.
- **Materialized CONs.**
  - Requires extra space.
  - Data may not be up-to-date.

Other Data Management Tasks

- How do we update, delete, or insert information in the system?
- **Ex: Add a person in the data.**

Materialization

Person	DOB
Amber A.	10/20/2030
Rose B.	10/20/1930
Dylan B.	**Unknown**
Deckard R.	**Unknown**

Virtualized

Name	DOB	Residency
Amber A.	10/20/2030	New Amsterdam
Rose B.	10/20/1930	Constantinople

ID	CName	Product
O001	Amber A.	P001
O002	Dylan B.	P001

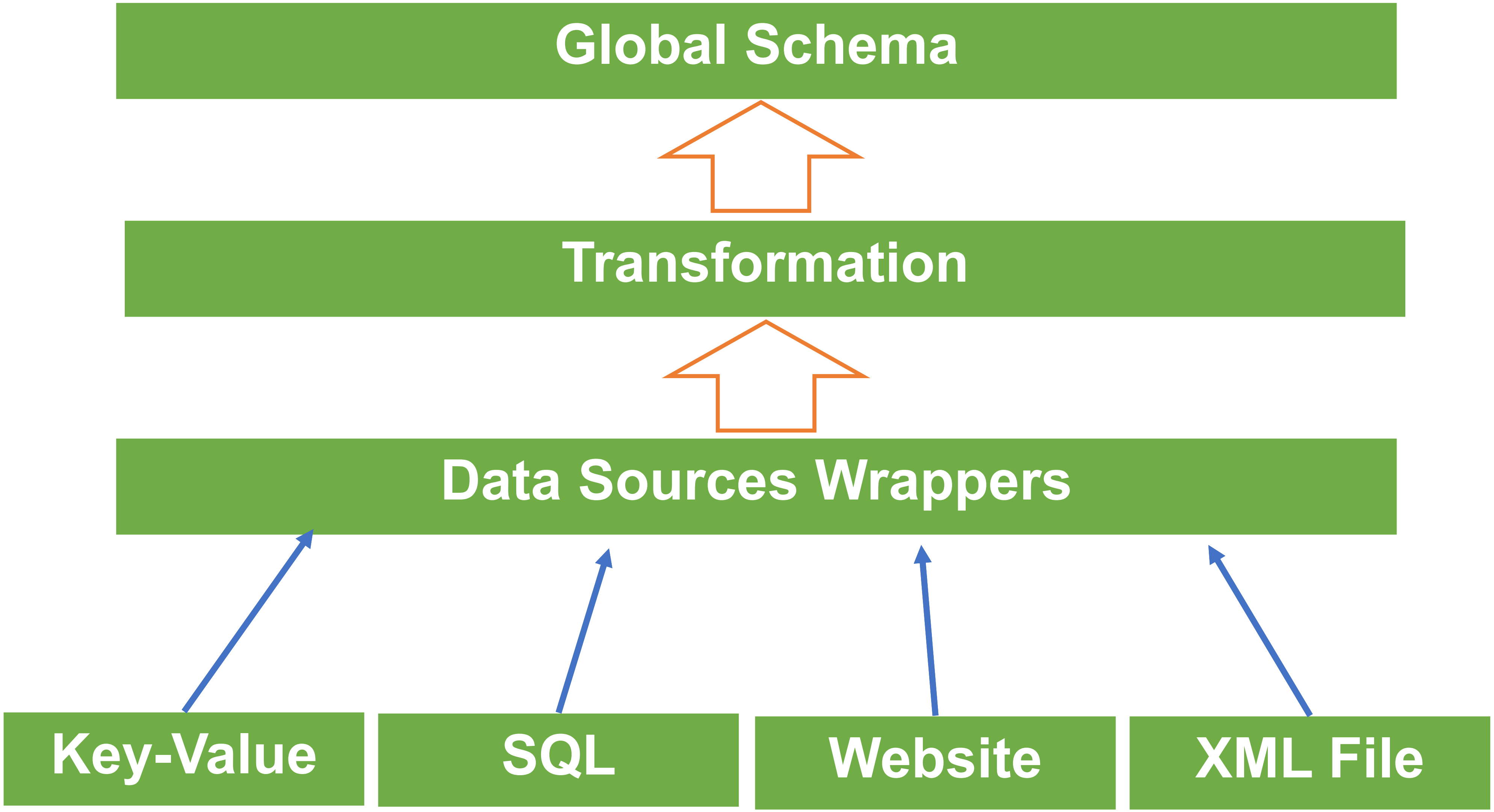
Deckard R.		

Solutions?

Dealing with Heterogeneity

- Physical Heterogeneity: **Wrappers**, i.e, pieces of software that extract data from the sources.
- Conceptual Heterogeneity: **Global Schema**, i.e., a conceptual description of the information we need.
- Result of the Integration: transformation from data at the sources to data for the global schema (mapping).

Dealing with Heterogeneity



Information Integration

Information integration is the problem of providing unified and transparent access to a collection of data stored in **multiple**, **autonomous**, and **heterogeneous** data sources.

Two different classes of approaches:

- **Materialization-based approaches**
  - A (small) representation of the result of the integration is loaded
  - Operations are carried out over the materialization
- **Virtualization-Based Approaches**
  - Tasks for the global schema are translated as queries for the data sources.

In both cases we need to define what the integration process means.

- We need to provide **semantics** for the specification

Challenges Ahead

- Information integration is (not only) a technological problem.

- We need to develop a **theory** to accommodate all its aspects.
  - How do we define a global schema?
  - How mapping reconcile the global schema with the sources?
  - How to answer queries expressed on the global schema?

In this part of the course, we aim to introduce the problem of **data integration** from a formal point of view and to develop such theory relying on formal tools such as

- The relational model of data
- Logic (Propositional Logic, First-Order Logic, and Description Logics)

Outline of the course

1. Introduction to Propositional Logic
2. Introduction to First-order Logic
3. Relational Calculus
4. **I**nformation **I**ntegration **S**ystems (IIS)
5. Logical formalization of IISs
6. Mapping between **Global Schema** e **Data Sources**
7. Incomplete information databases
8. Query answering over IISs
9. **O**ntology-**B**ased **D**ata **M**anagement (OBDM)
10. Description Logic Ontologies
11. Query answering in ontologies
12. Query answering in OBDM

Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.